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*The unit 1 is the universal basepoint: every number in every field is generated from it by the

3.2 The 90° axis as the algebraic-closure basepoint

Observation 3.2 (i is a y -axis choice). The imaginary unit i is the unit vector of the y -axis

Observation 3.3 (y -axis = basepoint for closure). The y -axis is a basepoint in the C007 sense

3.3 What a basepoint must carry to generate the number space

Design principle 3.4 (a generating basepoint). To generate the full human number-field space

1. **A unit** (the 1, the seed of successor and of every multiplication);
2. **An inverse operation** (yielding negatives and reciprocals);
3. **A completion/limit** (yielding reals, drawing ∞ into finite form);
4. **A perpendicular direction** (yielding ± 1 , closing the algebra).

The basepoint 1 with the operations of successor/inverse/reciprocal/limit/rotation satisfies all

4. Choosing the fastest-converging basepoint

4.1 Convergence defined

Definition 4.1 (axis peeling). *Axis peeling* is the removal of representation (projection-

Definition 4.2 (convergence). A structure *converges* when axis peeling reaches a fixed point

4.2 The fastest-convergence criterion

Principle 4.3 (fastest convergence). A basepoint converges fastest when it *exposes the generation*

- **∞ -axis**: choose the axis itself as the x' -axis ∞ coordinates are integers; the "irrational
- **Real line**: choose the circle (S^1) ∞ , e are structure constants; their transcendence is
- **NS**: choose $O(n)$ over $SO(n)$ ∞ time reversal T is a group element; the "irreversibility" is

Observation 4.4 (slow convergence). Choosing a basepoint that does not expose the generation

4.3 The convergence hierarchy as peeling order

Observation 4.5 (the number chain is the peeling sequence). The chain $\infty \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ is

5. No space remains at infinity

5.1 Compactification draws ∞ into the finite structure

Observation 5.1 (every space is compactified). Each level of the number chain is compactified

- **∞** : one-point compactification $\rightarrow S^1$ (the real line closes into a circle; $+\infty$ and $-\infty$ meet at
- **$\mathbb{R}$** : one-point compactification $\rightarrow S^2$ (the Riemann sphere; ∞ is the north pole).
- **The circle axis (S^1)**: on it, the structure constants ∞ and e are finite, intrinsic, and n

Definition 5.2 (not at infinity). A space is *not at infinity* when it is compactified: even

Observation 5.3 (peeling removes infinity). Axis peeling and compactification coincide: peel

6. The design recipe: a basepoint for the human number space

Design 6.1 (the recipe). To construct a basepoint that generates the full human number space

1. **Start from the unit 1** ∞ the seed of successor, inverse, reciprocal, limit, rotation.
2. **Add the inverse operation** (peel subtraction/division): ∞ , ∞ .
3. **Complete** (peel limits): ∞ ; compactify to S^1 (draw ∞ in).
4. **Take the 90° direction** (peel ± 1): ∞ ; compactify to S^2 (draw ∞ in).
5. **Choose the fastest-converging coordinates** at each level: the generating-structure-adapted

The resulting space: ∞ , algebraically closed, compact (Riemann sphere), with every number expressed

7. Conclusion

The human number-field space $\infty \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ is a convergence hierarchy under basepoint choice

Appendix A: identity checks (?????)

- $2 \cdot \cos 45^\circ = \sqrt{2}$ (verified: 1.414214); $2 \cdot \cos 30^\circ = \sqrt{3}$ (verified: 1.732051)
- $e^{(i \cdot 2\pi k)} = 1$ for all $k \in \mathbb{Z}$ (verified); $e^{(i \cdot (2k+1)\pi)} = -1$ for all $k \in \mathbb{Z}$ (verified)